

A Study of Hierarchical Neural Network Using Pulse Hardware Chaos Neuron Model for Character Image Recognition

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Research Background and Motivation

Brain-Inspired Computing

- Research on networks capable of mimicking biological processing of external information
- Focus on replicating chaotic spiking waveforms observed in biological neurons
- Implementation using standard CMOS processes for hardware realization

Challenges in Previous Research

- Teacher signals were applied to both preceding and post synaptic neurons
- Forcibly induced firing with artificial time differences
- Learning did not rely on natural timing of preceding and post synaptic firing

Spike Timing Dependent Plasticity (STDP)

- Biologically plausible learning algorithm attracting significant attention
- Adjusts synaptic weights based on timing differences between firing of preceding and post neurons

Research Motivation

- Develop a neural network that enables learning using asymmetric STDP circuit
- Eliminate need for manually inputting timing differences
- Create a system capable of natural learning based on intrinsic neural dynamics

Research Objectives

Primary Goal

Achieve Natural Learning in Hierarchical Neural Networks Using Asymmetric STDP Circuits



Architecture Design

Examine hierarchical neural network architecture that enables learning using asymmetric STDP circuit without manual timing input



Feedback Mechanism

Introduce excitatory synaptic feedback connections between preceding and post synaptic stages for natural timing-based learning



Practical Application

Enable learning, recall, and classification of character image data through natural neural dynamics

Expected Outcomes

 Demonstrate learning without manually inputting timing

 Successful classification of 9×9 character image data

Structure of Hierarchical Neural Network (H-NN)

P-HCNM (NM)

Pulse-type Hardware Chaos Neuron Model that replicates chaotic spiking waveforms observed in biological neurons

- Input layer: NM1-NM81 (9×9 pixels)
- Output layer: NM_[A][F][K]
- No hidden layer in this structure

Synapse Model (SyM)

Constructed using Operational Transconductance Amplifiers (OTAs)

- Used in recall network
- Receives STDP output voltage as input
- Fully connects input to output layer

STDP Circuit

Asymmetric type Spike-Timing-Dependent Plasticity circuit for learning

- Used in learning network

H-NN Structure Overview

Input layer (9×9 pixels)

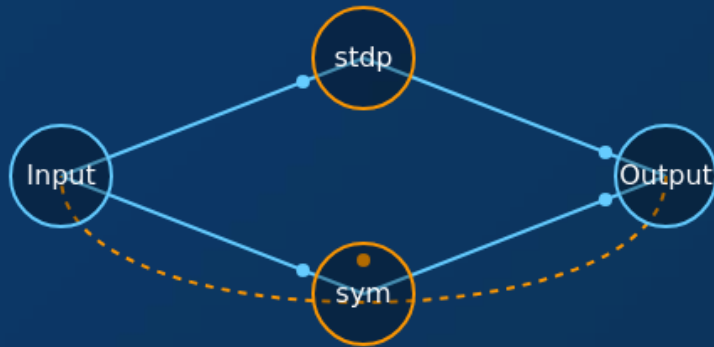


Key Structural Features:

- ✓ Separate learning and recall networks
- ✓ Full connectivity between layers
- ✓ Learning: STDP circuits modify weights
- ✓ Recall: SyMs use learned weights

Learning Network for a Single Pixel

Single Pixel Network Structure



Excitatory Feedback Connection

Connection Details

- ➔ Input layer P-HCNM connects to output layer P-HCNM via STDP circuits and synapse models
- ← Excitatory synapse model provides feedback connection from output layer to input layer
- ↔ Learning network is separated from recall network in overall H-NN structure

Feedback Mechanism

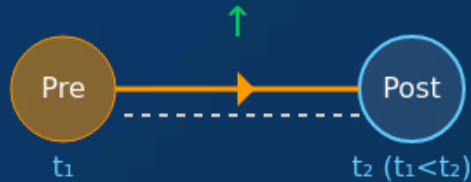
- » When input layer P-HCNM fires first:
 - Output layer P-HCNM is excited
 - Input layer P-HCNM enters refractory state
 - Feedback becomes ineffective
 - Synaptic weight increases via STDP
- « When output layer P-HCNM fires first:
 - If input layer P-HCNM is in excitable state, it fires
 - Synaptic weight decreases via STDP

Operating Principles of Feedback Mechanism

Feedback Mechanism for STDP Learning

The excitatory feedback connection between preceding and post-synaptic neurons enables natural STDP learning without manually imposed time differences.

↑ Synaptic Weight Strengthening



- ✓ When preceding P-HCNM fires **first**, causing post P-HCNM to fire
- ✓ Preceding neuron enters **refractory state**, making feedback ineffective
- ✓ STDP circuit increases output voltage from initial 0.9V
- ✓ Result: **Synaptic potentiation** (weight strengthening)

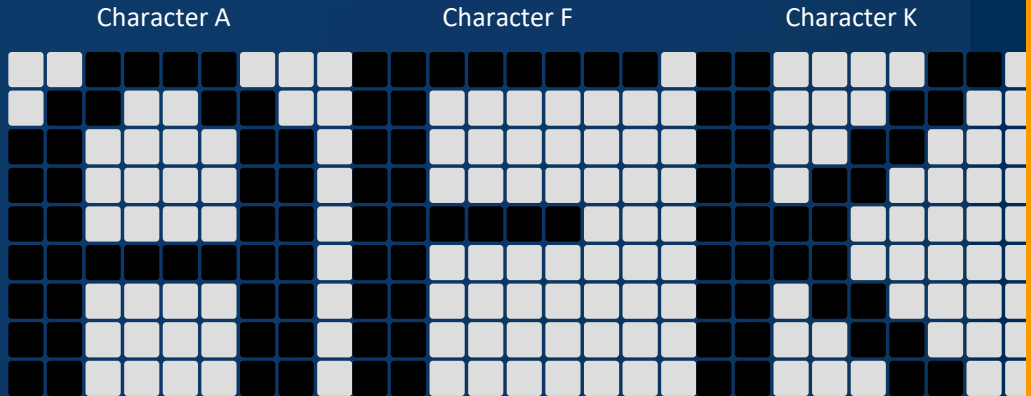
↓ Synaptic Weight Weakening



- ✓ Post P-HCNM fires **independently first**
- ✓ Feedback connection triggers preceding P-HCNM to fire
- ✓ STDP circuit decreases output voltage from initial 0.9V
- ✓ Result: **Synaptic depression** (weight weakening)

Experimental Setup

A Character Image Data

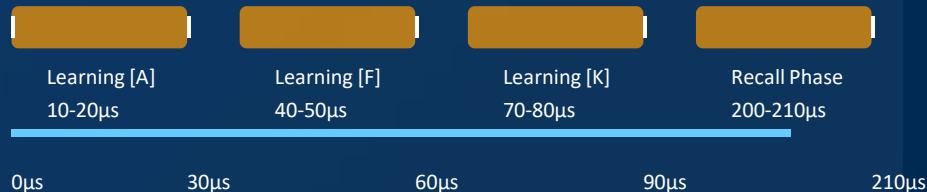


- 9×9 pixel binary images used for training and recognition
- Black pixels correspond to teacher signal application points

Learning Phase Setup

- **Teacher Signal:** Pulse waveform with amplitude of 18 μA and width of 10 μs
- **Bias Signal:** Pulse waveform with amplitude of 10 μA and width of 10 μs
- **STDP Initial Voltage:** All circuits set to 0.9 V
- **Learning Periods:**
 - Character [A]: 10 μs to 20 μs
 - Character [F]: 40 μs to 50 μs
 - Character [K]: 70 μs to 80 μs
- **Signal Application:** Teacher signal applied to P-HCNMs corresponding to black regions of character image

Timing Sequence



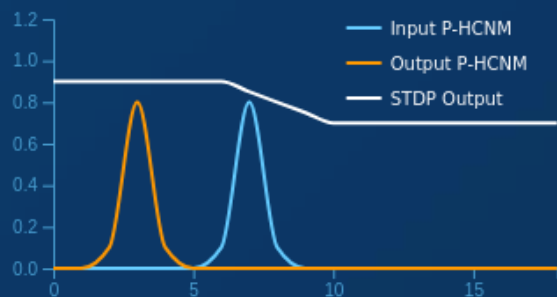
Recall Phase Setup

- **Recall Timing:** Character image data input between 200 μs and 210 μs
- **Input Signal:** Pulse waveform with amplitude of 18 μA and width of 10 μs

Learning Network Simulation Results

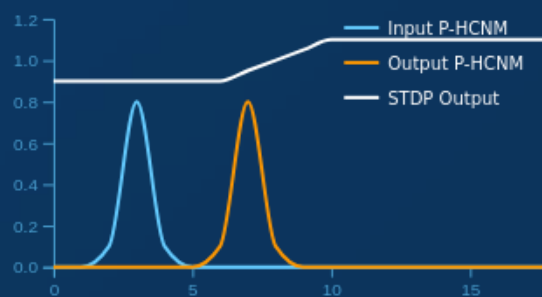
i Simulation results showing output waveforms of P-HCNMs and STDP circuits during character learning (pixel 01)

[A] Learning



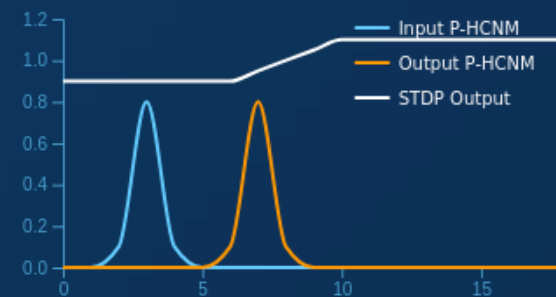
- No teacher signal applied to input layer P-HCNM
- Output layer P-HCNM fires first, followed by input layer P-HCNM (via feedback)
- ⚡ STDP output voltage **decreases** from initial 0.9V
- ↓ Result: Synaptic weight **weakens**

[F] Learning



- Teacher signal applied to input layer P-HCNM
- Input layer P-HCNM fires first, followed by output layer P-HCNM
- ⚡ STDP output voltage **increases** from initial 0.9V
- ↑ Result: Synaptic weight **strengthens**

[K] Learning

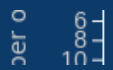


- Teacher signal applied to input layer P-HCNM
- Input layer P-HCNM fires first, followed by output layer P-HCNM
- ⚡ STDP output voltage **increases** from initial 0.9V
- ↑ Result: Synaptic weight **strengthens**

✓ Key Findings

The results demonstrate that learning using asymmetric STDP can be achieved by utilizing feedback from the output layer. Synaptic weights

Recall Network Simulation Results



11

Number of Firings in Output Layer P-HCNM for Different Input Characters

Firing Analysis

- > When character **[A]** is input: Only P-HCNM [A] fires (**11 times**)
- > When character **[F]** is input: Only P-HCNM [F] fires (**11 times**)
- > When character **[K]** is input: Only P-HCNM [K] fires (**5 times**)

Recognition Results

- > The P-HCNM with the highest firing probability corresponds to the input character
- > All three characters ([A], [F], [K]) were correctly recognized
- > This confirms successful recall operation of the learned character patterns

Conclusion

The simulation results demonstrate that learning in the hierarchical neural network (H-NN) using STDP is possible by employing the proposed feedback circuit. The network successfully classifies input character data based on the highest firing probability of the corresponding output layer P-HCNM.

✓ Conclusion

- Successfully proposed a configuration that enables learning using *asymmetric STDP* for the H-NN employing P-HCNMs
- Demonstrated learning and recall operations for *three 9×9 character image data* (A, F, K)
- Confirmed that synaptic weights can be:
 - Strengthened when post P-HCNM firing is caused by preceding P-HCNM
 - Weakened when preceding P-HCNM firing is caused by post P-HCNM
- Achieved successful recall and classification of learned character image data through feedback connections

🚀 Future Work

- Construct an *intermediate layer* capable of feature extraction
- Develop a network capable of learning and recalling *similar characters*
- Enhance the feedback mechanism to improve learning efficiency
- Explore hardware implementation challenges for practical applications



Our approach demonstrates potential for brain-inspired computing with natural learning dynamics in hardware neural networks

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Thank You